SuffAI Command Center — Deployment Guide

Step-by-step deploy for **role-based login ACL** and related Command Center changes (Geo / Ads Lab / Style Assistant / Admin). Canonical repo folder: `AI_CUP_latest`.

> **Audience:** operators redeploying an existing GCP stack
> **Region (default):** `asia-south1`
> **Primary service to redeploy:** Cloud Run `suffai-command-center`
> **New data:** BigQuery table `suffai_ops.login_cred_db`
> **Date:** October 2026

1. What these changes include

- BigQuery-backed login (`POST /v1/auth/login`, `/v1/auth/me`, `/v1/auth/logout`)
- Roles: **brand**, **customer**, **admin** with tab access control
- Brand users see own brand + **Competitors** on Geo/Ads Lab
- Customer users locked to their own Style Assistant wardrobe
- Seed credentials generator + BQ load script (no signup UI)

Access matrix

Role	Geo	Ads Lab	Style Assistant	Admin
brand	Yes (self + Competitors)	Yes (brand locked)	No	No
customer	No	No	Yes (own customer_id)	No
admin	Yes (all brands)	Yes	Yes	Yes

Services — what to redeploy

Cloud Run service	Redeploy?	Why
`suffai-command-center`	**Yes — required**	Auth API, UI ACL, metrics masking, Style/Ads enforcement
`suffai-extraction`	No	Unchanged for ACL
`suffai-similarity`	Only if you have other pending similarity changes	Not required for login ACL alone

> **No new secrets.** Auth uses the existing Cloud Run service account / ADC for BigQuery. Passwords in BQ are SHA-256 hex hashes. Plaintext passwords stay in a local gitignored CSV for handoff only.

Step 0 Prerequisites

1. Machine with Docker running, `gcloud` CLI, and `bq` CLI.
2. Access to the GCP project that already hosts Command Center.

3. Working directory:

```
cd /Users/rahul.kumar/Downloads/AI_CUP/AI_CUP_latest
```

```
gcloud auth login
gcloud auth configure-docker asia-south1-docker.pkg.dev

export GOOGLE_CLOUD_PROJECT=YOUR_PROJECT_ID   # replace
export GCP_REGION=asia-south1
gcloud config set project "$GOOGLE_CLOUD_PROJECT"
gcloud config set run/region "$GCP_REGION"
```

Confirm live services:

```
gcloud run services list --region="$GCP_REGION" --project="$GOOGLE_CLOUD_PROJECT" \
  --format='table(metadata.name,status.url)'
```

You should see at least `suffai-command-center`, `suffai-extraction`, `suffai-similarity`.

Capture URLs (needed for deploy env):

```
export EXTRACTION_URL=$(gcloud run services describe suffai-extraction \
  --region="$GCP_REGION" --project="$GOOGLE_CLOUD_PROJECT" \
  --format='value(status.url)')

export SIMILARITY_URL=$(gcloud run services describe suffai-similarity \
  --region="$GCP_REGION" --project="$GOOGLE_CLOUD_PROJECT" \
  --format='value(status.url)')

export CC_URL=$(gcloud run services describe suffai-command-center \
  --region="$GCP_REGION" --project="$GOOGLE_CLOUD_PROJECT" \
  --format='value(status.url)')

echo "EXTRACTION_URL=$EXTRACTION_URL"
echo "SIMILARITY_URL=$SIMILARITY_URL"
echo "CC_URL=$CC_URL"
```

Step 1 Generate login credentials (local)

Creates hashed CSV for BigQuery and a plaintext handoff file for you to distribute offline.

```
python3 command_center/scripts/generate_login_creds.py
```

Outputs under `command_center_data/auth/`:

File	Purpose
`login_cred_db.csv`	Upload to BigQuery (hashes only)
`login_creds_plaintext.csv`	**Handoff only** — username + clear password (gitignored)
`brand_logins_summary.csv`	Quick brand username summary

Do **not** commit `login_creds_plaintext.csv`. Keep it offline. Default admin seed: username `admin1`, password `123@admin`. Brand users look like `brand_nike`; customers like `cust_<id8>`.

Step 2 Load credentials into BigQuery

Creates (if needed) and loads `suffai_ops.login_cred_db`. Do this **before** users try the new login UI in production.

```
chmod +x command_center/scripts/load_login_creds_bq.sh
./command_center/scripts/load_login_creds_bq.sh "$GOOGLE_CLOUD_PROJECT"
```

Or pass an explicit CSV path:

```
./command_center/scripts/load_login_creds_bq.sh "$GOOGLE_CLOUD_PROJECT" \
  ./command_center_data/auth/login_cred_db.csv
```

Verify load

```
bq query --use_legacy_sql=false \
  "SELECT role, COUNT(*) AS n
   FROM \`${GOOGLE_CLOUD_PROJECT}.suffai_ops.login_cred_db\`
   GROUP BY role
   ORDER BY role"
```

Expect roughly: `admin` = 1, `brand` = 8, `customer` = 70.

Cloud Run SA for Command Center needs BigQuery Job User + Data Viewer (or equivalent) on dataset `suffai_ops` — same as existing Geo metrics. No new IAM roles beyond that for auth.

Step 3 (Optional) Cloud SQL migration 005

Only if Ads Lab campaign / Style Assistant tables are not already applied in this environment.

```
# If DATABASE_URL works locally:
psql "$DATABASE_URL" -v ON_ERROR_STOP=1 -f db/migrations/005_brand_sku_campaigns.sql

# Or use Cloud SQL Studio and paste the SQL file contents.
```

Skip this step if `brand_campaigns` / `brand_skus` / `brand_sku_pin_scores` already exist.

Step 4 Redeploy Command Center

Recommended path (resolves extraction/similarity URLs, builds, pushes, deploys):

```
chmod +x deploy/deploy_command_center.sh deploy/04_integrate_command_center.sh
./deploy/04_integrate_command_center.sh
```

Or explicitly:

```
export EXTRACTION_URL SIMILARITY_URL   # from Step 0
export GCS_IMAGE_BUCKET="${GCS_IMAGE_BUCKET:-${GOOGLE_CLOUD_PROJECT}-suffai-garment-images}"
export ENABLE_VERTEX=true
export VERTEX_LOCATION=asia-south1
export VERTEX_CHAT_MODEL=gemini-2.0-flash

./deploy/deploy_command_center.sh
```

Cloud Build alternative

```
gcloud builds submit --config=deploy/cloudbuild-command-center.yaml .

IMG="${GCP_REGION}-docker.pkg.dev/${GOOGLE_CLOUD_PROJECT}/suffai/suffai-command-center:latest"
gcloud run deploy suffai-command-center \
  --image="$IMG" \
  --region="$GCP_REGION" \
  --project="$GOOGLE_CLOUD_PROJECT" \
  --allow-unauthenticated \
  --memory=1Gi \
  --update-env-
vars="USE_BQ=true,BQ_DATASET=suffai_ops,GOOGLE_CLOUD_PROJECT=${GOOGLE_CLOUD_PROJECT},GCS_IMAGE_
BUCKET=${GCS_IMAGE_BUCKET},EXTRACTION_URL=${EXTRACTION_URL},SIMILARITY_URL=${SIMILARITY_URL},EN
ABLE_VERTEX=true,VERTEX_LOCATION=${VERTEX_LOCATION},VERTEX_CHAT_MODEL=${VERTEX_CHAT_MODEL},CORS
_ORIGINS=*"
```

Deploy scripts use `--update-env-vars` so existing keys (e.g. `DATABASE_URL` from Secret Manager) are preserved when not overwritten. After deploy, confirm Cloud SQL attachment still present if you use Style/Ads campaign persistence.

IAM checks (usually already set)

- CC SA → `roles/run.invoker` on `suffai-extraction` and `suffai-similarity`
- CC SA → BigQuery access on `suffai_ops` (metrics + `login_cred_db`)
- CC SA → Vertex AI User (chat + Style copy)

Step 5 Verify deployment

5.1 Health

```
CC_URL=$(gcloud run services describe suffai-command-center \
  --region="$GCP_REGION" --project="$GOOGLE_CLOUD_PROJECT" \
  --format='value(status.url)')

curl -sS "$CC_URL/health" | python3 -m json.tool
```

Expect flags including `use_bq`, `auth`, `ads_lab_campaign`, `style_assistant` true (as configured).

5.2 Auth login

```
curl -sS -X POST "$CC_URL/v1/auth/login" \
  -H 'Content-Type: application/json' \
  -d '{"username":"admin1","password":"123@admin"}' | python3 -m json.tool
```

Expect JSON with `token`, `role` = `admin`, and `allowed_tabs` including `geo`, `ads`, `style`, `admin`.

5.3 Metrics

```
curl -sS "$CC_URL/v1/metrics" | head -c 400; echo
```

5.4 UI smoke checklist

Open `$CC_URL/login.html` and verify:

□ **admin1** — all four tabs; full brand lists on Geo/Ads

□ **brand_*** (from plaintext CSV) — only Geo + Ads Lab; brand filter = own + Competitors; Ads brand field locked

□ **cust_*** — only Style Assistant; customer dropdown locked

□ Deep-link to a forbidden page (e.g. customer opening `geo.html`) redirects away

□ Corner Bot visible on Geo/Ads for brand/admin; hidden for customer / Style / Admin

□ Log out clears session and returns to login

Step 6 Distribute credentials

1. Open `command_center_data/auth/login_creds_plaintext.csv` locally.
2. Share brand rows with brand stakeholders; customer rows with demo customers; keep admin private.
3. Do not upload the plaintext file to GCS, Git, or Slack public channels.

Troubleshooting

Symptom	Likely cause	Fix
Login 401 Invalid username/password	BQ table empty / wrong hash / CSV not loaded	Re-run Step 1–2; verify `bq query` counts
Login 503 Auth unavailable	BQ IAM or project unset on Cloud Run	Check `GOOGLE_CLOUD_PROJECT`, SA BigQuery roles, Cloud Run logs
Tabs missing after login	Old UI revision / cached static HTML	Hard refresh; confirm new Cloud Run revision
Brand sees all brands	Metrics called without session token	Ensure redeployed UI sends auth headers; re-login
Sessions disappear after deploy	In-memory sessions cleared on new revision	Expected — users sign in again

Logs

```
gcloud run services logs read suffai-command-center \
  --region="$GCP_REGION" --project="$GOOGLE_CLOUD_PROJECT" --limit=50
```

Look for `[command_center]` and `auth login ok` lines.

Recommended order (short)

```
0. gcloud auth + export PROJECT / REGION / EXTRACTION_URL / SIMILARITY_URL
1. python3 command_center/scripts/generate_login_creds.py
2. ./command_center/scripts/load_login_creds_bq.sh "$GOOGLE_CLOUD_PROJECT"
3. ./deploy/04_integrate_command_center.sh
4. curl health + auth login + open /login.html
5. Distribute plaintext handoff offline
```

Reference docs in repo

- `docs/GCP_COMMAND_CENTER.md` — APIs, ACL matrix, BQ load notes
- `docs/GCP_REDEPLOY_E2E.md` — full end-to-end redeploy (campaign, Style, Bot)
- `command_center/scripts/generate_login_creds.py` — credential seed
- `command_center/scripts/load_login_creds_bq.sh` — BQ load

SuffAI / AI_CUP_latest — Deploy guide for role-based login ACL. Sessions are in-memory on Cloud Run (TTL ~12h); revison restarts clear active sessions. Out of scope: signup UI, Identity Platform, password reset.